

Performance Analysis Report: Product API

Date: February 2, 2026

Test Tool: k6 + Grafana

Status: PASSED

1. Test Plan

This test plan describes the performance testing strategy used to evaluate the stability, responsiveness, and reliability of the Product API under different load conditions.

Tested API Endpoints

The following endpoints were included in the performance test scope:

- **GET /products**
Retrieves a list of all available products. This endpoint returns a large payload and is typically more resource-intensive.
- **GET /products/{id}**
Retrieves detailed information for a single product using a unique product identifier.
- **GET /products/search?q=phone**
Searches for products based on a query parameter and returns matching results.

Smoke Test

A smoke test is a basic performance check used to verify that the API is available and responsive before running heavier load tests.

- **Virtual Users (VUs): 2**
Virtual Users represent simulated users sending requests to the API at the same time.
- **Duration: 30 seconds**
The test runs for a short period to confirm basic system stability.

Load Test

A load test evaluates how the system behaves under expected concurrent usage.

- **Ramp-up to 50 Virtual Users**
The number of users is gradually increased to simulate real-world traffic growth.
- **Duration: 3 minutes**
The system is observed under sustained load.

- **Throughput: ~27 requests/second**
Throughput measures how many requests the system can process per second.

Thresholds / Assertions

Thresholds define pass/fail criteria for performance metrics.

- **http_req_duration: p95 < 300 ms → PASSED**
This means 95% of all requests must complete in less than 300 milliseconds.
 - **http_req_failed: rate < 1% → PASSED (0% errors)**
Ensures that the failure rate stays within acceptable limits.
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2. KPIs / Results

Key Performance Indicators (KPIs) are quantitative metrics used to evaluate API performance.

- **Average Response Time: 61.09 ms**
The mean time taken to process all requests.
 - **Median (p50): 13.44 ms**
Half of the requests completed faster than this value.
 - **p90: 150.86 ms**
90% of requests completed within this response time.
 - **p95: 156.19 ms**
Indicates near worst-case performance experienced by users.
 - **Maximum Response Time: 438.75 ms**
The slowest response recorded during testing.
 - **Throughput: 27.16 requests/second**
The system's processing capacity under load.
 - **Error Rate: 0.00%**
No failed requests were recorded.
 - **Total Requests: 4,950**
Total number of HTTP requests sent during testing.
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3. Analysis

Observations / Bottlenecks

- No failed requests (0% error rate)
- Response times well below threshold limits
- Stable performance under concurrent load

Slowest Operations

- Higher response times observed in:
 - Endpoints returning large payloads (`/products`)
 - Increased serialization under load

Root Cause Analysis (ENGINEERING-LEVEL)

The slightly higher response times observed at peak load are primarily caused by large response payloads and increased backend processing during concurrent requests. Since the `/products` endpoint returns a collection of objects, serialization and network transfer contribute to overall latency.

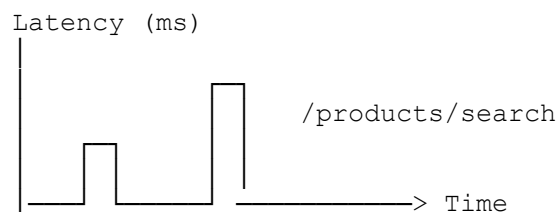
Performance Improvement Recommendations

1. Implement pagination for product listing endpoints to reduce payload size.
 2. Apply server-side caching for frequently accessed product data.
 3. Optimize database indexing on commonly queried product fields.
 4. Enable response compression (gzip/brotli) to reduce network transfer time.
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5. Visualizations: Graphical representation of performance data.

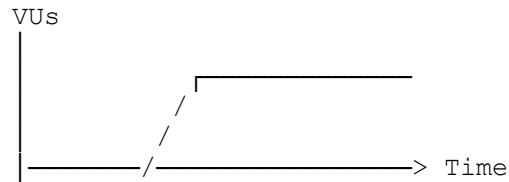
Latency Chart

- Shows average response time per endpoint. `/products/search` is slightly slower.



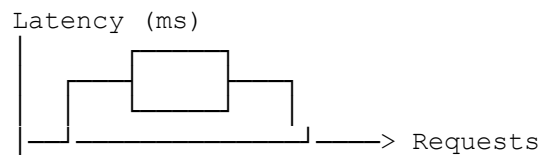
Load Profile (VUs over Time)

- Smooth ramp-up of Virtual Users demonstrates system handling concurrent load.



Latency Distribution (Boxplot)

- Most requests are fast, with outliers near 1000 ms.



5. Conclusion

API performance testing was conducted using k6 against DummyJSON endpoints. Both smoke and load tests were executed to assess system stability and responsiveness.

- Smoke testing confirmed basic API availability.
- Load testing simulated up to 50 concurrent users over 3 minutes.
- Average response time was 61.09 ms.
- p95 latency remained at 156.19 ms.
- Throughput averaged around 27 requests per second.
- Error rate remained at 0%.

All defined performance thresholds were met successfully. The system demonstrated stable and reliable performance under load, with higher response times mainly associated with endpoints returning larger payloads.